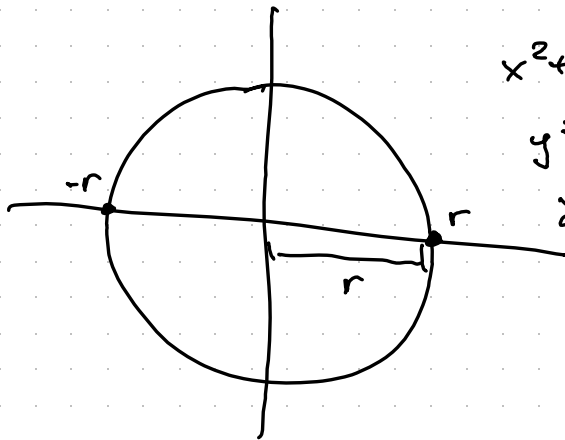


Sine substitution — 3.3

We saw that seeing $\sqrt{a^2 - x^2}$ in the integrand often suggests the trig sub $x = a \sin \theta$.

Problem: What is the area of a circle of radius r ?



$$x^2 + y^2 = r^2$$

$$y^2 = r^2 - x^2$$

$$y = \pm \sqrt{r^2 - x^2}$$

$$\text{Area} = 2 \int_{-r}^r \sqrt{r^2 - x^2} dx$$

$$2 \int_{-r}^r \sqrt{r^2 - x^2} dx$$

$$x = r \sin \theta$$

$$dx = r \cos \theta d\theta$$

$$r = r \sin \theta$$

$$1 = \sin \theta$$

$$\theta = \frac{\pi}{2}$$

$$-r = r \sin \theta$$

$$-1 = \sin \theta$$

$$\theta = -\frac{\pi}{2}$$

$$= 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{r^2 - r^2 \sin^2 \theta} \cdot r \cos \theta d\theta$$

$$= 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{r^2 \cos^2 \theta} \cdot r \cos \theta d\theta = 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} r \cos \theta \cdot r \cos \theta d\theta$$

$$= 2r^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 \theta d\theta = 2r^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1 + \cos(2\theta)}{2} d\theta$$

$$= 2r^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} d\theta + 2r^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos(2\theta)}{2} d\theta$$

$$= 2r^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} d\theta + 2r^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos(2\theta)}{2} d\theta$$

$$= r^2 \theta \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + r^2 \cdot \frac{1}{2} \sin(2\theta) \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$= r^2 \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) + \frac{1}{2} r^2 \left(\cancel{\sin(\pi)}^0 - \cancel{\sin(-\pi)}^0 \right)$$

$$= r^2 \left(\frac{\pi}{2} + \frac{\pi}{2} \right) = \pi r^2.$$

Example: Find $\int \frac{\sqrt{4-x^2}}{x} dx$ $x = 2\sin\theta$
 $dx = 2\cos\theta d\theta$

$$= \int \frac{\sqrt{4-4\sin^2\theta}}{2\sin\theta} \cdot 2\cos\theta d\theta = \int \frac{2\cos\theta}{2\sin\theta} \cdot 2\cos\theta d\theta$$

$$= 2 \int \frac{\cos^2\theta}{\sin\theta} d\theta = 2 \int \frac{1-\sin^2\theta}{\sin\theta} d\theta$$

$$= 2 \int \frac{1}{\sin\theta} d\theta - 2 \int \sin\theta d\theta$$

$$= 2 \int \csc\theta d\theta + 2\cos\theta + C$$

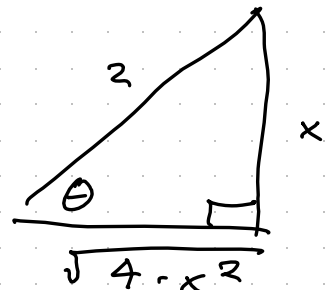
$$= 2 \ln|\csc\theta - \cot\theta| + 2\cos\theta + C$$

$$= 2 \ln |\csc \theta - \cot \theta| + 2 \cos \theta + C$$

$$= 2 \ln \left| \frac{2}{x} - \frac{\sqrt{4-x^2}}{x} \right| + \sqrt{4-x^2} + C$$

$$= 2 \ln \left| \frac{2 - \sqrt{4-x^2}}{x} \right| + \sqrt{4-x^2} + C$$

$$\frac{x}{2} = \sin \theta$$



$$\cos \theta = \frac{\sqrt{4-x^2}}{2}$$

$$\cot \theta = \frac{\sqrt{4-x^2}}{x}$$

$$\csc \theta = \frac{2}{x}$$